David Vivar Bogónez

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SUMMARY

Computer Engineering undergraduate at the Universidad de Castilla-La Mancha (UCLM) and open-source developer. Technical focus on systems programming, networking, developer tooling, and Linux, with primary development experience in Rust and Python. Native Spanish speaker with Cambridge English C1 certification.

EDUCATION

Universidad de Castilla-La Mancha (UCLM) Ciudad Real, Spain
BSc in Computer Engineering, Escuela Superior de Ingeniería Informática (ESI) 2025 – Present

TECHNICAL SKILLS

Languages: Python, Rust, Java/Scala, SQL, JavaScript/TypeScript
Databases: PostgreSQL, SQLite
Tools and platforms: Linux, Docker/Podman, Git and Github, Nix
Areas: Systems programming, network programming, concurrency, databases, operating systems, agentic programming

PROJECTS

artscii | *Rust* artscii.pages.dev
- Rust workspace for image and video to ASCII conversion with a CLI, reusable library crates, and a WebAssembly web app.
- Supports multiple dithering algorithms, ANSI, HTML, GIF, and MP4 output, with reproducible Nix packaging.

sentinel | *Python*
- Python CLI for process supervision with process groups, log rotation, HTTP/TCP health checks, and persistent state.
- Provides real-time CPU, memory, and uptime monitoring, JSON/table metrics export, remote SSH management, and systemd unit generation.

http | *Rust*
- Built the reusable `http` and `websocket` Rust crates and a demo server, implementing HTTP/1.1 and RFC 6455 from scratch on Tokio.
- Includes RFC compliance matrices, architecture and threat-model documentation, fuzzed parsers, CI regression tests, and Docker Compose offensive demos.

bfcompiler | *Rust* bfcompiler.pages.dev
- Rust workspace with a Brainfuck compiler, resumable interpreter, terminal REPL, and WebAssembly playground.
- Compiles Brainfuck to Rust source and native executables, with instruction budgets to pause non-terminating programs.

perlin | *Scala*
- Scala 3 engine for infinite procedural terrain with layered Perlin and ridged noise, biomes, deterministic seeds, and parallel generation strategies.
- First-person software 3D renderer with chunk streaming plus PNG, ANSI, and SVG map export.

go-pane | *QML*
- Self-contained KDE Plasma 6 QML plasmoid for live OpenCode Go usage across rolling, weekly, and monthly windows.
- Calculates pace ratios, projected usage, and exhaustion estimates, with API key discovery and configurable warning thresholds.

Additional projects: pmatrix (Python), particles (TypeScript), py-logic (Python).

LANGUAGES

English: C1 Advanced (CAE) - Grade C, Score 192 (Cambridge English)
Spanish: Native